CDK12 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes CDK12 gene fusions from the msk_impact_50k_2026 study, covering 43 fusion events. 7/43 fusions (16.3%) were in-frame. 16/43 fusions (37.2%) retained the PF00069 domain. Domain retention was tested with Fisher's exact test (p=0.0677006, not statistically significant at alpha=0.05). No literature benchmark is configured for CDK12.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 43 analyzed fusion events, identifying 29 distinct fusion partner genes. The most frequent partner was FBXL20, observed in 6 events.

Domain retention

PF00069 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 5/7 (71.4%) of in-frame fusions retained the domain, p=0.0677006 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.316832.

Domain disruption

Domain-disruption analysis was skipped: CDK12 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 41 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 644 aa (permutation-corrected p=0.00990099, statistically significant at alpha=0.05). This is -84 aa from the nearest configured domain boundary at 728 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=16) and "not_retained" (n=25). For Frame_status, retained: 5/7 (71.4%, 95% CI 35.9%-91.8%); not_retained: 2/7 (28.6%, 95% CI 8.2%-64.1%). Welch's t-test comparing Tumor_variant_count between groups gave means of 20.8125 vs 43.4, p=0.0502833 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: CDK12 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ADCY9	2	0.046511627906976744	0.04991713954627366	None	1.0	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.2756851385529113	1
AKAP8L	1	0.023255813953488372	0.04991713954627366	None	0.9906291834002677	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.2652071802079033	2
HTR1E	2	0.046511627906976744	0.04991713954627366	None	0.9250334672021419	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.2616289136533129	3
MED13	1	0.023255813953488372	0.04991713954627366	None	0.9451137884872824	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.25667304366171856	4
ODAD4	1	0.023255813953488372	0.04991713954627366	None	0.9210174029451138	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.252154971372562	5
TBKBP1	1	0.023255813953488372	0.04991713954627366	None	0.9210174029451138	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.252154971372562	6
GRB7	1	0.023255813953488372	0.04991713954627366	None	0.85809906291834	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.24035778261754187	7
ERBB2	3	0.06976744186046512	0.04991713954627366	None	0.7367246764837126	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.23504194562616557	8
RECQL4	1	0.023255813953488372	0.04991713954627366	None	0.7817938420348058	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.22605055370187926	9
STAT5B	4	0.09302325581395349	0.04991713954627366	None	0.5917001338688086	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.21657077411842918	10
COLEC12	1	0.023255813953488372	0.04991713954627366	None	0.6974564926372155	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.21023730068983107	11

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CACNA1G	1	0.023255813953488372	0.04991713954627366	None	0.6613119143239625	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.20346019225609613	12
FBXL20	6	0.13953488372093023	0.04991713954627366	None	0.4152164212405176	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.20092193846574088	13
SYNRG	1	0.023255813953488372	0.04991713954627366	None	0.6265060240963856	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.19693408783842545	14
ANKFN1	1	0.023255813953488372	0.04991713954627366	None	0.6050870147255689	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.1929180235813973	15
SKAP1	1	0.023255813953488372	0.04991713954627366	None	0.5917001338688086	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.19040798342075477	16
USP6	1	0.023255813953488372	0.04991713954627366	None	0.5917001338688086	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.19040798342075477	17
CSF3	1	0.023255813953488372	0.04991713954627366	None	0.5394912985274432	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.18061882679424873	18
FAM222B	1	0.023255813953488372	0.04991713954627366	None	0.4029451137884873	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.15501641715569453	19
IKZF3	2	0.046511627906976744	0.04991713954627366	None	0.321285140562249	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.14842610240833298	20
SPON1	1	0.023255813953488372	0.04991713954627366	None	0.32797858099062915	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.1409601922560961	21
SEC31A	1	0.023255813953488372	0.04991713954627366	None	0.2449799196787149	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.1253979432601122	22
VMP1	1	0.023255813953488372	0.04991713954627366	None	0.2449799196787149	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.1253979432601122	23
SNF8	1	0.023255813953488372	0.04991713954627366	None	0.18340026773761708	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.11385175852115635	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
MAP3K3	2	0.046511627906976744	0.04991713954627366	None	None	0.441153375836676	['confidence_certainty', 'domain_retention', 'recurrence']	0.10853555514204467	25
MTRES1	1	0.023255813953488372	0.04991713954627366	None	0.048192771084337394	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.08850035289866641	26
IGFBP4	1	0.023255813953488372	0.04991713954627366	None	0.01740294511378848	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.08272726052918851	27
FGFR4	1	0.023255813953488372	0.04991713954627366	None	0.01338688085676043	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.08197424848099574	28
RPS6KB1	1	0.023255813953488372	0.04991713954627366	None	0.0	0.441153375836676	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.07946420832035316	29

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
80	1	15	0	25	None	0.39024390243902435	0.4086638740638108
240	2	14	0	25	None	0.14634146341463414	0.8346326063360919
246	3	13	0	25	None	0.052532833020637895	1.279569177684353
349	4	12	0	25	None	0.01797175866495507	1.7454094219943264
603	5	11	0	25	None	0.0058286784859313745	2.2344299000136965
637	6	10	0	25	None	0.0017809850929234755	2.749339715622759
644	9	7	0	25	None	3.265366098560993e-05	4.486068120461908
700	9	7	2	23	14.785714285714286	0.0011510086327457892	2.938921419081068
703	11	5	2	23	25.3	7.698377534159563e-05	4.113600794594795
750	11	5	3	22	16.133333333333333	0.0004274646628148892	3.369099781237133
807	11	5	4	21	11.55	0.0009913088581971262	3.003791013009668
870	11	5	5	20	8.8	0.003014843556286022	2.520735218981372
897	11	5	6	19	6.966666666666667	0.008600185164931898	2.0654921981468433
923	11	5	9	16	3.911111111111111	0.05784571299099636	1.2377288215582707
949	11	5	15	10	1.4666666666666666	0.7420848641039097	0.12954642638599925
955	11	5	16	9	1.2375	1.0	-0.0

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
988	11	5	17	8	1.035294117647059	1.0	-0.0
1032	12	4	17	8	1.411764705882353	0.7341702326198091	0.13420322820990868
1090	12	4	18	7	1.1666666666666667	1.0	-0.0
1121	12	4	19	6	0.9473684210526315	1.0	-0.0
1146	13	3	19	6	1.368421052631579	1.0	-0.0
1159	13	3	20	5	1.0833333333333333	1.0	-0.0
1208	13	3	22	3	0.5909090909090909	0.6623956829348356	0.17888250666352729
1254	15	1	22	3	2.0454545454545454	1.0	-0.0
1355	16	0	22	3	None	0.26829268292682923	0.5713911715615105
1378	16	0	23	2	None	0.5121951219512195	0.2905645619858162
1381	16	0	24	1	None	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

5	11
2	23

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	41	16	15	10	0.22484641638225256

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
4	19	43	0.4418604651162791

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ADCY9	2
AKAP8L	1

Partner_gene	Event_count
ANKFN1	1
CACNA1G	1
COLEC12	1
CSF3	1
ERBB2	3
FAM222B	1
FBXL20	6
FGFR4	1
GRB7	1
HTR1E	2
IGFBP4	1
IKZF3	2
MAP3K3	2
MED13	1
MTRES1	1
ODAD4	1
RECQL4	1
RPS6KB1	1
SEC31A	1
SKAP1	1
SNF8	1
SPON1	1
STAT5B	4
SYNRG	1
TBKBP1	1
USP6	1
VMP1	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039125-T01-IM6-1	P-0039125-T01-IM6		AKAP8L-CDK12	AKAP8L	unknown	three_prime	37627994	2	637	False	retained	1.0	False	Protein kinase domain			AKAP8L (NM_014371) - CDK12 (NM_016507) rearrangement: t(17;19)(q12;p13.12)(chr17:g.37627994::chr19:g.15529663) | Protein Fusion: mid-exon {AKAP8L:CDK12}	NA
EVT-P-0056622-T01-IM6-2	P-0056622-T01-IM6		ANKFN1-CDK12	ANKFN1	in-frame	three_prime	37619399	2	349	True	retained	1.0	False	Protein kinase domain			ANKFN1 (NM_153228) - CDK12 (NM_016507) rearrangement: c.62+8631:ANKFN1_c.1046+29:CDK12dup | Protein Fusion: in frame {ANKFN1:CDK12}	NA
EVT-P-0003762-T02-IM5-4	P-0003762-T02-IM5		CDK12-ADCY9	ADCY9	unknown	three_prime	37646665	3	644	True	retained	1.0	False	Protein kinase domain			ADCY9 (NM_001116) - CDK12 (NM_016507) rearrangement: t(16;17)(p13.3;q12)(chr16:g.4033430::chr17:g.37646665) | Protein fusion: mid-exon (ADCY9-CDK12)	NA
EVT-P-0003762-T01-IM5-5	P-0003762-T01-IM5		CDK12-ADCY9	ADCY9	unknown	three_prime	37646665	3	644	True	retained	1.0	False	Protein kinase domain			t(16;17)(p13.3;q12)(chr16:g.4033430::chr17:g.37646665) | Protein fusion: mid-exon (ADCY9-CDK12)	NA
EVT-P-0024871-T01-IM6-7	P-0024871-T01-IM6		CDK12-CACNA1G	CACNA1G	unknown	five_prime	37667806	8	897	False	disrupted	0.5802047781569966	True			Protein kinase domain	CDK12 (NM_016507) - CACNA1G (NM_018896) rearrangement: c.2691:CDK12_c.6399+226:CACNA1Gdel | Protein Fusion: mid-exon {CDK12:CACNA1G}	NA
EVT-P-0038990-T01-IM6-53	P-0038990-T01-IM6		CDK12-FBXL20	FBXL20	in-frame	five_prime	37676367	11	1032	True	retained	1.0	False	Protein kinase domain			CDK12 (NM_016507) - FBXL20 (NM_032875) rearrangement: c.3095+27:CDK12_c.104+18457:FBXL20inv | Protein Fusion: in frame {CDK12:FBXL20}	NA
EVT-P-0002174-T01-IM3-55	P-0002174-T01-IM3		CDK12-GRB7	GRB7	unknown	three_prime	37650768	5	750	True	disrupted	0.9249146757679181	True			Protein kinase domain	NA | Transcript fusion (GRB7-CDK12)	NA
EVT-P-0008960-T02-IM6-56	P-0008960-T02-IM6		CDK12-HTR1E	HTR1E	unknown	five_prime	37646977	3	700	False	lost	0.0	False		Protein kinase domain		CDK12 (NM_016507) - HTR1E (NM_000865) rearrangement: t(6;17)(q14.3;q12)(chr6:g.87697113::chr17:g.37646977) | Protein Fusion: mid-exon {CDK12:HTR1E}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008960-T01-IM5-57	P-0008960-T01-IM5		CDK12-HTR1E	HTR1E	unknown	five_prime	37646977	3	700	False	lost	0.0	False		Protein kinase domain		CDK12 (NM_016507) - HTR1E (NM_000865) rearrangement: t(6;17)(q14.3;q12)(chr6:g.87697112::chr17:g.37646977) | Protein fusion: mid-exon (CDK12-HTR1E)	NA
EVT-P-0060111-T01-IM7-58	P-0060111-T01-IM7		CDK12-IKZF3	IKZF3	in-frame	five_prime	37682742	13	1254	True	retained	1.0	False	Protein kinase domain			CDK12 (NM_016507) - IKZF3 (NM_012481) rearrangement: c.3760+173:CDK12_c.8-12056:IKZF3inv | Protein Fusion: in frame {CDK12:IKZF3}	NA
EVT-P-0040752-T01-IM6-62	P-0040752-T01-IM6		CDK12-MTRES1	MTRES1	unknown	five_prime	37687160	14	1355	False	retained	1.0	False	Protein kinase domain			CDK12 (NM_016507) - C6orf203 (NM_016487) rearrangement: t(6;17)(q21;q12)(chr6:g.107368450::chr17:g.37687160) | Protein Fusion: mid-exon {CDK12:C6orf203}	NA
EVT-P-0009835-T01-IM5-67	P-0009835-T01-IM5		CDK12-RECQL4	RECQL4	out-of-frame	five_prime	37656296	5	807	True	disrupted	0.27303754266211605	True			Protein kinase domain	CDK12 (NM_016507) - RECQL4 (NM_004260) rearrangement: t(8;17)(q24.3;q12)(chr8:g.145739493::chr17:g.37656296) | Protein fusion: out of frame (CDK12-RECQL4)	NA
EVT-P-0023550-T01-IM6-68	P-0023550-T01-IM6		CDK12-RPS6KB1	RPS6KB1	unknown	three_prime	37687269	14	1391	False	lost	0.0	False		Protein kinase domain		CDK12 (NM_016507) rearrangement: c.4173_chr17:g.58028299dup | Protein Fusion: mid-exon {RPS6KB1:CDK12}	NA
EVT-P-0041371-T01-IM6-69	P-0041371-T01-IM6		CDK12-SKAP1	SKAP1	out-of-frame	five_prime	37673680	9	949	True	disrupted	0.757679180887372	True			Protein kinase domain	CDK12 (NM_016507) - SKAP1 (NM_003726) rearrangement: c.2847-13:CDK12_c.280+66555:SKAP1inv | Protein Fusion: out of frame {CDK12:SKAP1}	NA
EVT-P-0043326-T01-IM6-71	P-0043326-T01-IM6		CDK12-SNF8	SNF8	out-of-frame	five_prime	37683411	13	1254	True	retained	1.0	False	Protein kinase domain			CDK12 (NM_016507) - SNF8 (NM_007241) rearrangement: c.3760+842:CDK12_c.422+1405:SNF8inv | Protein Fusion: out of frame {CDK12:SNF8}	NA
EVT-P-0057341-T01-IM6-73	P-0057341-T01-IM6		CDK12-SPON1	SPON1	unknown	five_prime	37682247	13	1146	False	retained	1.0	False	Protein kinase domain			CDK12 (NM_016507) - SPON1 (NM_006108) rearrangement: t(11;17)(p15.2;q12)(chr11:g.14149386::chr17:g.37682247) | Protein Fusion: mid-exon {CDK12:SPON1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016507-T04-IM6-74	P-0016507-T04-IM6		CDK12-STAT5B	STAT5B	unknown	three_prime	37672113	10	949	True	disrupted	0.24573378839590443	True			Protein kinase domain	CDK12 (NM_016507) - STAT5B (NM_012448) rearrangement: c.2846+52:CDK12_c.2238-225:STAT5Bdel | Antisense Fusion	NA
EVT-P-0016507-T03-IM6-75	P-0016507-T03-IM6		CDK12-STAT5B	STAT5B	unknown	three_prime	37672113	10	949	True	disrupted	0.24573378839590443	True			Protein kinase domain	CDK12 (NM_016507) - STAT5B (NM_012448) rearrangement: c.2846+52:CDK12_c.2238-225:STAT5Bdel. | Antisense Fusion	NA
EVT-P-0016507-T02-IM6-76	P-0016507-T02-IM6		CDK12-STAT5B	STAT5B	unknown	three_prime	37672113	10	949	True	disrupted	0.24573378839590443	True			Protein kinase domain	CDK12 (NM_016507) - STAT5B (NM_012448) rearrangement: c.2846+52:CDK12_c.2238-225:STAT5Bdel | Antisense Fusion	NA
EVT-P-0016507-T01-IM6-77	P-0016507-T01-IM6		CDK12-STAT5B	STAT5B	unknown	three_prime	37672113	10	949	True	disrupted	0.24573378839590443	True			Protein kinase domain	CDK12 (NM_016507) - STAT5B (NM_012448) rearrangement: c.2846+52:CDK12_c.2238-225:STAT5Bdel | Antisense Fusion	NA
EVT-P-0023516-T01-IM6-78	P-0023516-T01-IM6		COLEC12-CDK12	COLEC12	out-of-frame	three_prime	37665943	7	870	True	disrupted	0.515358361774744	True			Protein kinase domain	COLEC12 (NM_130386) - CDK12(NM_016507) rearrangement: t(17;18)(q12;p11.32)(chr17:g.37665943::chr18:g.386902) | Protein Fusion: out of frame {COLEC12:CDK12}	NA
EVT-P-0038406-T01-IM6-80	P-0038406-T01-IM6		CSF3-CDK12	CSF3	out-of-frame	three_prime	37676083	11	988	True	disrupted	0.11262798634812286	True			Protein kinase domain	CSF3 (NM_000759) - CDK12 (NM_016507) Rearrangement : c.313-58:CSF3_c.2964-126:CDK12dup | Protein Fusion: out of frame {CSF3:CDK12}	NA
EVT-P-0046790-T01-IM6-81	P-0046790-T01-IM6		ERBB2-CDK12	ERBB2	out-of-frame	three_prime	37640037	3	644	True	retained	1.0	False	Protein kinase domain			ERBB2 (NM_004448) - CDK12 (NM_016507) rearrangement: c.225+207:ERBB2_c.1932-6773:CDK12dup | Protein Fusion: out of frame {ERBB2:CDK12}	NA
EVT-P-0043706-T01-IM6-82	P-0043706-T01-IM6		ERBB2-CDK12	ERBB2	unknown	three_prime	37668120	9	923	True	disrupted	0.33447098976109213	True			Protein kinase domain	ERBB2 (NM_004448) - CDK12 (NM_016507) rearrangement: c.2293:ERBB2_c.2768+237:CDK12dup | Protein Fusion: mid-exon {ERBB2:CDK12}	NA
EVT-P-0037167-T01-IM6-83	P-0037167-T01-IM6		ERBB2-CDK12	ERBB2	unknown	three_prime	37673710	10	955	False	disrupted	0.22525597269624573	True			Protein kinase domain	ERBB2 (NM_004448) - CDK12 (NM_016507) rearrangement: c.2553:ERBB2_c.2864:CDK12dup | Protein Fusion: mid-exon {ERBB2:CDK12}	NA

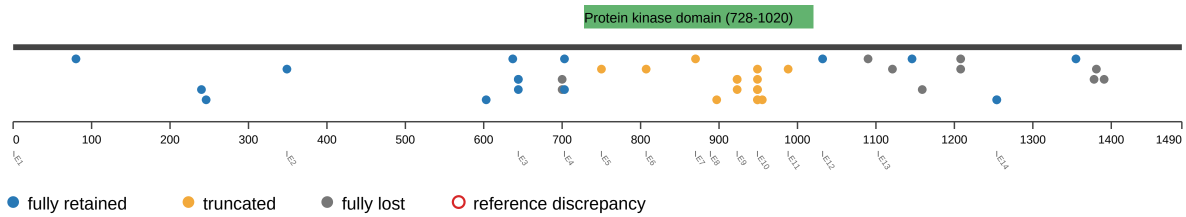
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036357-T01-IM6-84	P-0036357-T01-IM6		FAM222B-CDK12	FAM222B	unknown	three_prime	37681100	12	1090	False	lost	0.0	False		Protein kinase domain		CDK12 (NM_016507) rearrangement: c.3269:CDK12_chr17:g.27096697inv | Protein Fusion: mid-exon {FAM222B:CDK12}	NA
EVT-P-0014403-T01-IM6-86	P-0014403-T01-IM6		FBXL20-CDK12	FBXL20	unknown	three_prime	37682171	13	1121	False	lost	0.0	False		Protein kinase domain		FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.43-9338:FBXL20_c.3362:CDK12inv | Protein fusion: mid-exon (FBXL20-CDK12)	NA
EVT-P-0046675-T01-IM6-87	P-0046675-T01-IM6		FBXL20-CDK12	FBXL20	in-frame	three_prime	37671717	9	923	True	disrupted	0.33447098976109213	True			Protein kinase domain	FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.398+4667:FBXL20_c.2769-267:CDK12inv | Protein Fusion: in frame {FBXL20:CDK12}	NA
EVT-P-0043910-T01-IM6-88	P-0043910-T01-IM6		FBXL20-CDK12	FBXL20	unknown	three_prime	37682285	13	1159	False	lost	0.0	False		Protein kinase domain		FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+20635:FBXL20_c.3476:CDK12inv | Protein Fusion: mid-exon {FBXL20:CDK12}	NA
EVT-P-0036553-T01-IM6-89	P-0036553-T01-IM6		FBXL20-CDK12	FBXL20	unknown	three_prime	37619062	1	246	False	retained	1.0	False	Protein kinase domain			FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.697-2344:FBXL20_c.738:CDK12inv | Protein Fusion: mid-exon {FBXL20:CDK12}	NA
EVT-P-0048892-T01-IM6-90	P-0048892-T01-IM6		FBXL20-CDK12	FBXL20	unknown	three_prime	37682432	13	1208	False	lost	0.0	False		Protein kinase domain		FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+25196:FBXL20_c.3623:CDK12inv | Protein Fusion: mid-exon {FBXL20:CDK12}	NA
EVT-P-0026288-T03-IM6-91	P-0026288-T03-IM6		FGFR4-CDK12	FGFR4	unknown	three_prime	37687238	14	1381	False	lost	0.0	False		Protein kinase domain		FGFR4 (NM_213647) - CDK12 (NM_016507) rearrangement: t(5,17)(q35.2;q12)(chr5:g.176518227::chr17:g.37687238) | Protein Fusion: mid-exon {FGFR4:CDK12}	NA
EVT-P-0060015-T01-IM7-92	P-0060015-T01-IM7		IGFBP4-CDK12	IGFBP4	unknown	three_prime	37687228	14	1378	False	lost	0.0	False		Protein kinase domain		IGFBP4 (NM_001552) - CDK12 (NM_016507) rearrangement: c.349+3777:IGFBP4_c.4132:CDK12dup | Protein Fusion: mid-exon {IGFBP4:CDK12}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0068695-T01-IM7-93	P-0068695-T01-IM7		IKZF3-CDK12	IKZF3	unknown	three_prime	37619043	1	240	False	retained	1.0	False	Protein kinase domain			IKZF3 (NM_012481) - CDK12 (NM_016507) rearrangement: c.8-3620:IKZF3_c.719:CDK12inv | Protein Fusion: mid-exon {IKZF3:CDK12}	NA
EVT-P-0048104-T01-IM6-94	P-0048104-T01-IM6		MED13-CDK12	MED13	unknown	three_prime	37627894	2	603	False	retained	1.0	False	Protein kinase domain			MED13 (NM_005121) - CDK12 (NM_016507) rearrangement: c.2264-2554:MED13_c.1809:CDK12inv | Protein Fusion: mid-exon {MED13:CDK12}	NA
EVT-P-0065545-T01-IM7-95	P-0065545-T01-IM7		ODAD4-CDK12	ODAD4	in-frame	three_prime	37648612	4	703	True	retained	1.0	False	Protein kinase domain			TTC25 (NM_031421) - CDK12 (NM_016507) rearrangement: c.1059-1499:TTC25_c.2109-392:CDK12dup | Protein Fusion: in frame {TTC25:CDK12}	NA
EVT-P-0048892-T01-IM6-97	P-0048892-T01-IM6		SEC31A-CDK12	SEC31A	unknown	three_prime	37682432	13	1208	False	lost	0.0	False		Protein kinase domain		SEC31A (NM_014933) - CDK12 (NM_016507) rearrangement: t(4;17)(q21.22;q12)(chr4:g.83761890::chr17:g.37682432) | Protein Fusion: mid-exon {SEC31A:CDK12}	NA
EVT-P-0005530-T02-IM5-98	P-0005530-T02-IM5		SYNRG-CDK12	SYNRG	out-of-frame	three_prime	37671800	9	923	True	disrupted	0.33447098976109213	True			Protein kinase domain	SYNRG (NM_007247) - CDK12 (NM_016507) rearrangement : c.118+961:SYNRG_c.2769-184:CDK12inv | Protein fusion: out of frame (SYNRG-CDK12)	NA
EVT-P-0041371-T01-IM6-99	P-0041371-T01-IM6		TBKBP1-CDK12	TBKBP1	in-frame	three_prime	37648975	4	703	True	retained	1.0	False	Protein kinase domain			TBKBP1 (NM_014726) - CDK12 (NM_016507) rearrangement: c.872+1376:TBKBP1_c.2109-29:CDK12dup | Protein Fusion: in frame {TBKBP1:CDK12}	NA
EVT-P-0019776-T01-IM6-100	P-0019776-T01-IM6		USP6-CDK12	USP6	in-frame	three_prime	37673592	10	949	True	disrupted	0.24573378839590443	True			Protein kinase domain	USP6 (NM_004505) - CDK12 (NM_016507) rearrangement: c.2916-470:USP6_c.2847-101:CDK12del | Protein Fusion: in frame {USP6:CDK12}	NA
EVT-P-0059578-T01-IM7-101	P-0059578-T01-IM7		VMP1-CDK12	VMP1	unknown	three_prime	37618564	1	80	False	retained	1.0	False	Protein kinase domain			VMP1 (NM_030938) - CDK12 (NM_016507) rearrangement: c.582+1747:VMP1_c.240:CDK12dup | Protein Fusion: mid-exon {VMP1:CDK12}	NA

Figures

domain retention outliers

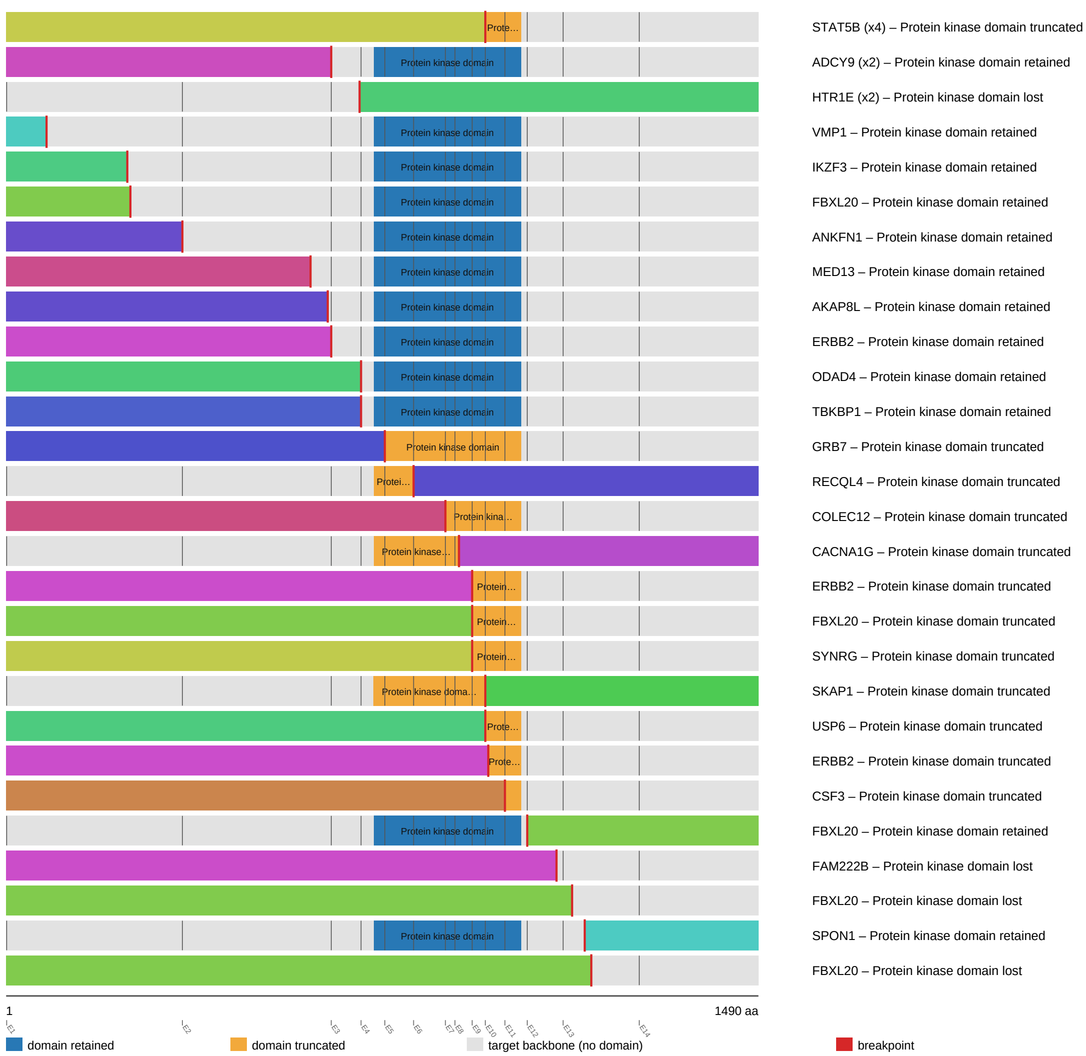
CDK12 domain-retention track



fusion schematic

CDK12 fusion-transcript schematic

partner block → breakpoint → retained CDK12 portion (domain-colored, exon ticks)



Showing the top 28 of 36 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

reference comparison

Reference vs msk_impact_50k_2026